

cat3 Brian

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ABERDARE TECHNICAL COLLEGE  
P. O. BOX 1815 - 10101 KARATINA TEL: 0726507444  
E-MAIL: INSTITUTE.ABERDARE@GMAIL.COM

APPLY BASIC ELECTRONICS

INSTRUCTIONS TO CANDIDATE

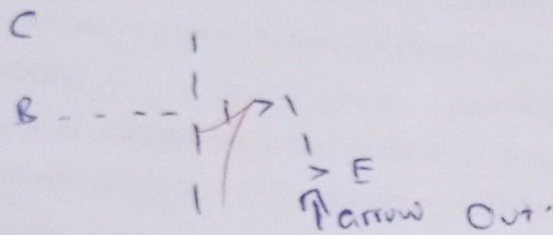
1. This paper consists of **TWO** sections: **A** and **B**.
2. Attempt **ALL** questions in section A and **ANY THREE** (3) questions in section B.
3. Marks for each question are indicated in the brackets.
4. Do not write on the question paper.

## Section A.

### 1] Difference b/w NPN & PNP Bipolar Transistors

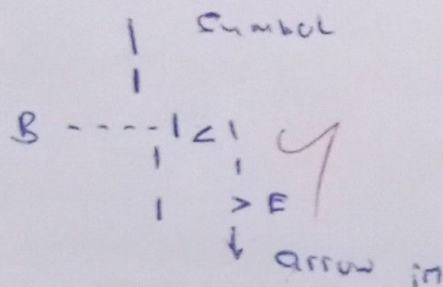
#### NPN Transistor

- ① Majority Charge Carriers are Electrons
  - ② Turns ON when the Base is more + than the Emitter
  - ③ Current flows from Collector to emitter.
- Symbol NPN



#### PNP Transistor

- Majority Charge Carriers are holes
- Turns ON when the Base is more negative than the Emitter.
- Current flows from Emitter to Collector.



### Semiconductor memory

Stores data electronically using integrated circuits

Very fast but usually expensive

### Optical memory

Stores data using laser light on optical discs.

Slower, but suitable for long-term storage.

### Ammeter Reading.

3] Given:

- Two Cells in Parallel
- $\text{Emf} = 1.5\text{V}$
- internal Resistance  $= 0.5\Omega$

Two Bulbs of  $6\Omega$  Each in Series.

Total Resistance  $= 6 + 6 = 12\Omega$

Total Circuit Resistance  $= 12 + 0.5 = 12.5\Omega$

$$\text{Current} = I = \frac{V}{R} = \frac{1.5}{12.5} = 0.12\text{A}$$

$$= 0.12\text{A}$$

4] a) Signal Diode

- allow current to flow in one direction
- used for rectification and signal switching.

Capacitor

- Stores Electrical Charge.
- used for filtering, timing and smoothing voltage.

5] Conductor  $\rightarrow$  Conductivity decreases as temperature increases

Semiconductors  $\rightarrow$  Conductivity increases as temperature increases.

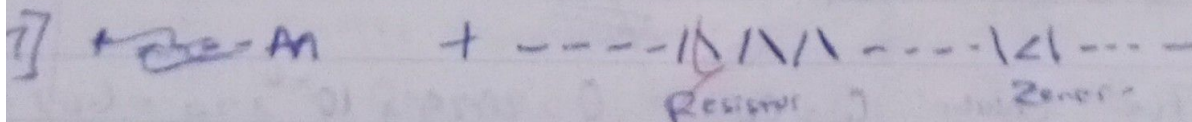
6]  $135_{10}$  to Binary  $\rightarrow 10000111_2$

$126_{10}$  to Hexadecimal  $\rightarrow 7E_{16}$

$101101_2$  to decimal  $\rightarrow 45_{10}$

$1434_{10}$  to Binary  $\rightarrow 1100011100_2$

Show the working.

7]   
Resistor Zener

uses:

1] Voltage Regulation 2] voltage Protection against overvoltage

3] If  $Z_1$  turns off the circuit becomes open, so no current flows

Answer = 607 uV

1) When Surface  $\phi$  is  $\phi_1$  then  $\phi_2$  is  $\phi_1$   
 When Surface  $\phi$  is  $\phi_2$  then  $\phi_1$  is  $\phi_2$   
 Because both surfaces are  $\phi_1$  and  $\phi_2$

2) Volume Band

- Contains Electrons and  
 Holes and  $\phi_1$
- Electrons  $\phi_2$   
 Holes  $\phi_1$

Volume Band  
 Contains Electrons and  
 Holes  $\phi_1$  and  $\phi_2$   
 Electrons  $\phi_2$   
 Holes  $\phi_1$

1) From  $\phi_1$  to  $\phi_2$  (Electrons)

From  $\phi_2$  to  $\phi_1$  (Electrons)

From  $\phi_1$  to  $\phi_2$  (Holes)

2) From  $\phi_1$  to  $\phi_2$

- Loss Energy :  $0.20 \text{ eV}$  (Electrons)
- Gain Energy :  $0.25 \text{ eV}$
- Net Voltage :  $0.05 \text{ V}$

3) Population Calculations

$$C_1 = \frac{C_{e1}}{C_{e1} + C_{h1}} = \frac{0.20 \times 10^{21}}{0.20 \times 10^{21} + 0.25 \times 10^{21}} = 0.444$$

Since this is parallel with  $25 \text{ uV}$

$$V_T = 25 + 0.201 = 25.201 \text{ uV}$$

$$\text{Total Charge } Q = C \cdot V = 0.25 \times 10^{-12} \times 25.201 = 6.30 \times 10^{-12} \text{ C}$$

Answer =  $607 \text{ uV}$

High Immunity to  $\phi_1$  and  $\phi_2$

High Reliability, Cost Efficiency

In General

The depletion layer becomes  $\phi_1$  (Electrons)

- As Forward Voltage increases, Current increases rapidly because more charge carriers cross the junction.

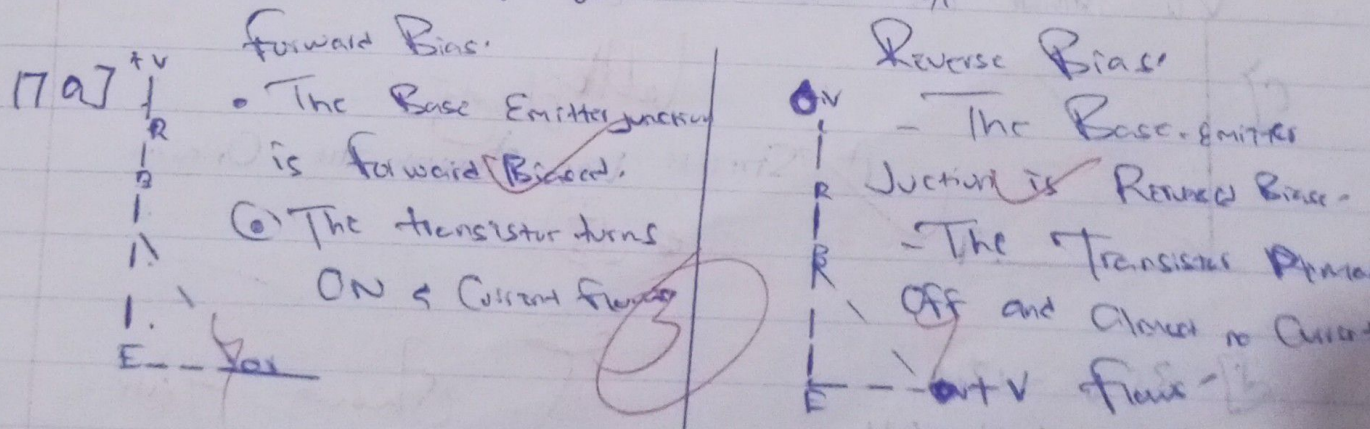
## Section E.

18. a) Both have Processor [CPU] that executes instructions.  
 - Both have memory [RAM & Storage] for programs & data.  
 Both use an Operating System to manage hardware and software.  
 > Both can connect to the internet using network - *explain!*  
 - Both can run applications such as browsers, games and office.

- b) Resistors  $\rightarrow$  Limit current and divide voltage.  
 Capacitors  $\rightarrow$  Stores electrical charge and smooth/filter signals.  
 Transistors  $\rightarrow$  Used for switching and amplification.

- c) Ransomware or production disruption  $\rightarrow$  malware can shut down manufacturing systems, causing delay, downtime and significant financial losses.

Intellectual property theft  $\rightarrow$  Hackers steal circuit designs, software and product information, leading to financial losses and loss of competitive advantage.

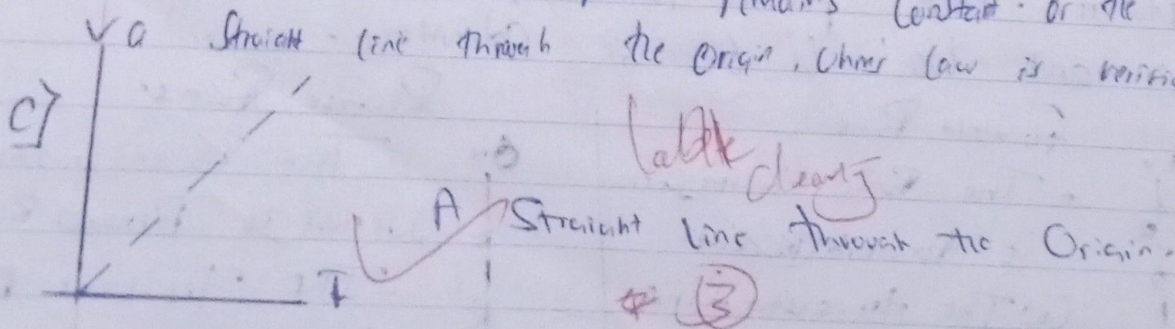


- b) High Conversion Cost  $\rightarrow$  Replacing analogue components with digital circuits requires new equipment and redesign which is expensive.

Compatibility Issue -  $\rightarrow$  Existing analogic devices may not work directly with digital systems, requiring interface or conversion circuit.

- 1) Carry out market Research to understand customer need.
- 2) Improve product quality and Reliability.
- 3) introduce portable and smart digital products.
- 4) invest in Research and development.
- 5) Use Competitive Pricing.
- 6) Improve After Sale Service and Warranties.
- 7) Increase advertising and digital marketing.
- 8) Train Staff on modern technologies.
- 9) Upgrade manufacturing processes.
- 10) Build Partnership with technology Companies.

- 16] a)  $\rightarrow$  ~~State~~ Ohm's law  $\rightarrow$  States that the Current flowing through a Conductor is directly proportional to the Voltage across it, provided the temperature and other physical constant remain const.  $V = IR$ .
- b) - close the Switch and use the rheostat to vary the Current. Record the Voltmeter and Ammeter readings for different current. Calculate  $V/I$  for each reading or plot a graph of  $V$  against Current. If the ratio remains constant or the graph is a straight line through the origin, Ohm's law is verified.



$$\begin{aligned} \sqrt{35}_{10} &= 17_{10} \\ 35 &= 00100011_2 \end{aligned}$$

$$17 = 00010001_2$$

$2^5$  Complement of 17

$$1^5 \text{ Complement} = 11101110 + 1 = 11101111$$

$$\text{Answer} = 00010010 = 18_{10}$$

Show your working here

using 1's Complement Subtract 17 from 28

$$28 = 00011100_2 \quad \checkmark$$

$$17 = 00010001_2 \quad \checkmark$$

1's Complement of 17

$$11011110$$

Add

$$\begin{array}{r} 00011100 \\ 11011110 \\ \hline 10001010 \quad \checkmark \end{array}$$

Add End Around Carry

$$0001010$$

+

$$00001011$$

$$= 11_2$$

~~28~~